

Extended Abstract

Motivation Despite recent advances in alignment techniques such as Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO), instruction-following models still exhibit inconsistent behavior. Small variations in decoding—introduced by sampling randomness—can lead to drastically different outputs, ranging from helpful and aligned to incoherent or incorrect. Compounding this issue, the reward models used to evaluate or guide model behavior (e.g., Nemotron-70B) often reflect underspecified or noisy preferences. These dual sources of variability make it difficult to reliably select the best response. In this work, we ask whether leveraging inference-time compute—by sampling multiple completions and systematically selecting among them—can improve the robustness of instruction-following without any additional finetuning.

Method Our approach consists of three stages. First, we fine-tune the Qwen2.5-0.5B base model on Smol-SmolTalk, a dataset designed for small models, to build strong instruction-following behavior using Supervised Fine-Tuning (SFT). Next, we apply Direct Preference Optimization (DPO) using the UltraFeedback Binarized dataset, aligning the model with human preferences via a contrastive learning objective on completion pairs. Finally, we propose an inference-time extension that improves robustness by sampling multiple completions per prompt and reranking them using external reward models. This extension introduces no additional training cost and enables test-time robustness improvements by leveraging available inference-time compute.

Implementation We fine-tuned and evaluated the Qwen2.5-0.5B model using two open-source datasets: Smol-SmolTalk and UltraFeedback Binarized. Smol-SmolTalk contains 460k synthetic dialogues with a 24.2k-example test split and was used for Supervised Fine-Tuning (SFT). UltraFeedback Binarized contains over 60k preference-labeled prompts and a 2k-example test split, and was used for DPO as well as evaluation. Final comparisons were performed on 400 held-out prompts provided by the course staff. All training was conducted on a single A100 GPU with a 256-token input and 1024-token output length, batch size 8, and AdamW optimizer. For baseline evaluations (SFT and DPO), we generated completions using vLLM with temperature = 0.6, top-k = 20, top-p = 0.95, and max tokens = 1024. For inference-time scaling experiments, we varied decoding settings and sampled $N \in \{1, 4, 8, 16\}$ completions per prompt. Each was scored using Nemotron-70B and GPT4o-mini based reward models, with z-score normalization applied before selection. We additionally used cross-verifier agreement and a fallback-to-greedy mechanism when reward signal variance was low. Evaluation metrics included win-rate (Nemotron), and qualitative model output inspection.

Results We evaluate our models on 400 held-out prompts provided by the course staff. Starting from a DPO-trained baseline with a 61% win-rate (measured using the Nemotron-70B reward model against the base Qwen model), our inference-time reranking strategy boosts performance to approximately 68–70% when selecting the best of 8 sampled completions. Gains taper off beyond $N=8$, suggesting diminishing returns. We find that reward model choice substantially affects outcomes: GPT4o-mini rubric scoring outperforms Nemotron across all configurations, indicating better alignment with desirable responses. Normalizing scores with z-scoring improves stability, and our fallback mechanism mitigates decision noise in low-confidence cases. Together, these findings demonstrate that inference-time reranking offers consistent quality improvements without modifying model parameters.

Discussion & Conclusion Our results suggest that inference-time reranking can meaningfully improve the reliability of small instruction-following models without retraining. By sampling multiple completions and selecting among them with external reward models, we mitigate the effects of both decoding randomness and reward signal noise. This approach is practical in deployment settings where additional compute is available at inference but retraining is infeasible. However, our reliance on automated reward models introduces potential bias and misalignment with real user preferences. Future work should explore human evaluation, multilingual robustness, and more adaptive reranking strategies, such as learned or generative verifiers. Overall, our findings highlight inference-time reranking as a lightweight, effective method for improving instruction-following quality in practice.

Scaling Inference-Time Compute for Robust Instruction-Following

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Abstract

Instruction-following language models often generate inconsistent or suboptimal outputs due to stochastic decoding and underspecified reward functions. While techniques like Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) improve alignment during training, they do not fully address variability at inference time. We propose a lightweight extension that improves model robustness by scaling inference-time compute: sampling multiple completions and selecting the best using external reward models. We apply this method to a 0.5B parameter model trained on Smol-SmolTalk and UltraFeedback Binarized, and evaluate on 400 held-out prompts. Our approach improves win-rate by up to 9 percentage points over a DPO-only baseline, with performance peaking at 8 sampled completions. These results suggest that test-time reranking is a practical, retraining-free strategy for improving instruction-following quality in deployment settings.

1 Introduction

Instruction-following is a core capability for language models (LMs) deployed in assistant and alignment-critical settings. Recent techniques such as Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) have significantly improved model behavior by aligning outputs with human preferences. Yet, even well-aligned models can produce erratic outputs at inference time due to the inherent randomness of decoding processes and the limitations of reward signals used to guide model behavior.

Specifically, stochastic sampling often leads to completions that vary widely in fluency, helpfulness, or coherence—even for the same input. Reward models, while useful for supervision or post hoc evaluation, are often noisy, underspecified, or brittle, introducing further uncertainty into output quality. These sources of variability pose a challenge for real-world applications where reliability and robustness are critical.

This project investigates whether scaling inference-time compute—by generating multiple candidate completions per prompt and systematically selecting among them—can improve instruction-following quality without additional training. Unlike contrastive decoding or beam search, our approach is model-agnostic and applicable to any decoder-only language model without modifying architecture or loss functions.

We make the following contributions:

- We fine-tune Qwen2.5-0.5B on Smol-SmolTalk and apply DPO using UltraFeedback Binarized to align with human preferences.

- We introduce an inference-time reranking strategy that selects from multiple sampled completions using external reward models, including Nemotron-70B and GPT4o-mini rubric scoring.
- We conduct a detailed analysis of decoding diversity, normalization strategies, and verifier agreement, using 400 held-out evaluation prompts.
- Our approach yields a 7–9% improvement in reward-model win-rate over the DPO-only baseline, demonstrating the effectiveness of inference-time compute scaling for robustness.

2 Related Work

Alignment via Supervised and Preference-Based Learning. Supervised Fine-Tuning (SFT) on curated instruction-following datasets has become a standard approach for aligning large language models (LLMs) with human intent [Ouyang et al. (2022a); Chung et al. (2022)]. Preference-based learning techniques, such as Reinforcement Learning from Human Feedback (RLHF) [Ouyang et al. (2022b)] and Direct Preference Optimization (DPO) [Rafailov et al. (2023)], further refine alignment by leveraging ranked model completions. While these methods improve average output quality, they do not address the variance introduced by sampling at inference time.

Reward Modeling and Evaluation. Reward models play a central role in alignment pipelines by providing scalar feedback used for finetuning or evaluation. Recent datasets such as UltraFeedback [Cui et al. (2023)] enable high-quality reward modeling by pairing prompts with completions scored by GPT4o-mini across several axes (e.g., helpfulness, honesty). Models such as Nemotron-70B [Wang et al. (2024)] and OpenAI’s GPT4o-mini rubric have been used as reward functions in recent alignment work [Xu et al. (2023)]. However, reward models can be biased, brittle, or underspecified, and often disagree with human preferences on edge cases [Wu (2025)].

Inference-Time Sampling and Reranking. Various decoding strategies aim to improve inference-time output quality. Sampling-based methods (e.g., top- k , nucleus sampling [Holtzman et al. (2020)]) increase response diversity, while deterministic strategies like beam search reduce stochasticity but risk low diversity. More advanced methods such as contrastive decoding [Li et al. (2023)] and self-consistency sampling [Wang et al. (2022)] have shown promise in reasoning tasks. Recent work in math and code generation has explored reranking multiple completions using reward signals [Madaan et al. (2023)]. However, these strategies remain underexplored in open-ended instruction-following, where the outputs are more linguistically varied and the reward models less precise.

Our Contribution. We extend inference-time reranking to the domain of instruction-following by systematically evaluating best-of- N sampling with external reward models. Our approach is modular, does not require retraining, and highlights the sensitivity of output quality to verifier choice, normalization, and sampling configuration.

3 Method

Our approach consists of three sequential stages: Supervised Fine-Tuning (SFT), Direct Preference Optimization (DPO), and an inference-time reranking extension. This pipeline is designed to produce a compact instruction-following model that is robust to decoding variability and reward signal noise without requiring retraining for each new inference-time improvement.

3.1 Supervised Fine-Tuning on Smol-SmolTalk

We begin by training a decoder-only language model, Qwen2.5-0.5B, on the Smol-SmolTalk dataset. This dataset comprises over 460,000 English-language dialogues filtered for small-scale instruction-following tasks and designed specifically for models under 1B parameters. The conversations are mostly single-turn or short multi-turn exchanges, covering a broad range of general-purpose instructions. During SFT, the model is trained to predict assistant responses given a user prompt. To make training on the Smol-SmolTalk dataset tractable for the Qwen2.5-0.5B model, we truncated each multi-turn dialogue to just the final user/assistant message pair. We applied the Qwen chat template manually and inserted EOS tokens, which the model does not add automatically — omitting EOS

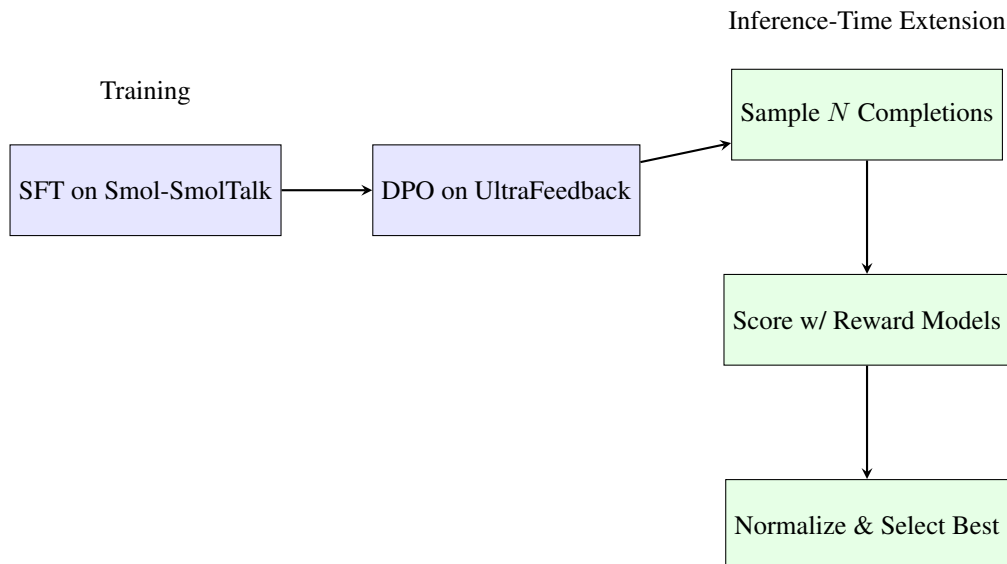


Figure 1: Method Overview. We train a small language model with SFT and DPO, and at inference time generate N completions, score them using external reward models (Nemotron-70B or GPT4o-mini), normalize, and select the highest-ranked response.

severely degraded training loss and output quality. We limited the maximum output sequence length to 1024 tokens to reduce memory usage, and ultimately used standard SDPA attention after encountering a Qwen-specific error when attempting to use FlashAttention (stemming from a tokenizer bug with left/right padding modes). Based on prior recommendations, we preprocessed each input into three strings: s_1 , s_2 , and s_3 , which were used to create the model’s input and target. We also manually constructed the attention loss mask to only supervise assistant tokens, due to an incorrect default mask behavior in the Qwen tokenizer that incorrectly labels all tokens with 1.

3.2 Direct Preference Optimization on UltraFeedback

To further align the model with human preferences, we apply Direct Preference Optimization (DPO) using UltraFeedback Binarized. This dataset provides prompt pairs with "chosen" and "rejected" completions, derived from GPT4o-mini preference scores. DPO optimizes a contrastive objective based on the log-likelihood difference between the preferred and dispreferred completions. Unlike RLHF, DPO does not require training a separate reward model and remains stable under simple KL-regularized fine-tuning. This step improves alignment beyond SFT by directly incorporating ranked feedback.

3.3 Inference-Time Reranking via Reward Models

Finally, we introduce an inference-time extension that improves robustness by leveraging additional compute without modifying the model weights. For each prompt, we sample multiple completions using diverse decoding configurations and rank them using external reward models. We explore both Nemotron-70B—a high-capacity reward model trained on UltraFeedback—and GPT4o-mini rubric-based scoring, which is more general-purpose. To mitigate reward noise, we normalize scores (e.g., via z-scoring), apply fallback mechanisms when score variance is low, and analyze cross-verifier agreement between Nemotron and GPT4o-mini. The highest-scoring completion is returned as the final output.

This “horizontal search” method allows the model to hedge against both decoding variance and misaligned reward signals, improving robustness without retraining. It is compatible with any base model and can be deployed as a lightweight inference-time wrapper.

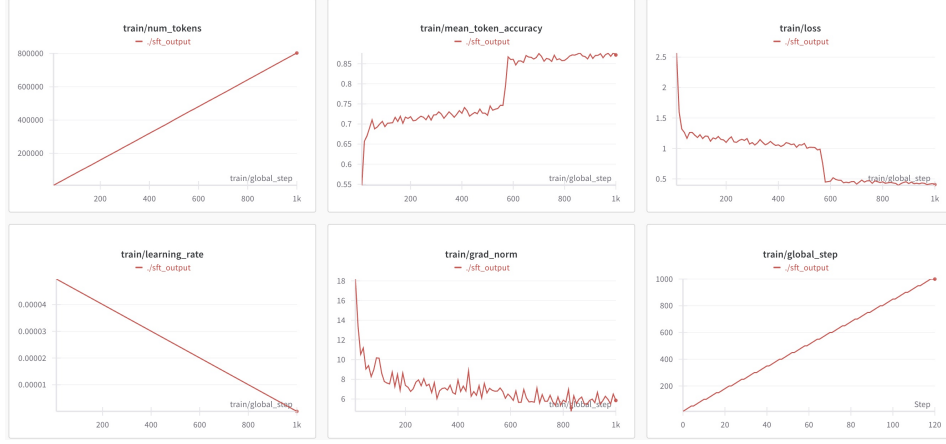


Figure 2: Training loss curve for SFT on Smol-SmolTalk



Figure 3: Training loss curve for DPO on UltraFeedback.

4 Experimental Setup

4.1 Model and Training Procedure

We use the Qwen2.5-0.5B base model, a decoder-only transformer with approximately 500 million parameters. All fine-tuning was performed using the Hugging Face Transformers and Pytorch libraries on a single NVIDIA A100 GPU with 40GB of memory. We used a maximum input length of 256 tokens and allowed for up to 1024 output tokens during generation. Training was conducted with a batch size of 8 and mixed-precision (bfloat16) enabled for memory efficiency.

4.2 Datasets

Smol-SmolTalk. This dataset contains over 460,000 English-language dialogues tailored to instruction-following in small models. Compared to the original SmolTalk dataset, Smol-SmolTalk includes shorter, simpler conversations and excludes function-calling or advanced reasoning tasks. We use the training split for SFT and the 24.2k example test split for evaluating SFT-only model performance.

UltraFeedback Binarized. UltraFeedback is a large preference dataset where each prompt is paired with multiple completions labeled by GPT4o-mini. UltraFeedback Binarized converts this into a binary format by selecting the top-ranked completion as the "chosen" response and randomly sampling a "rejected" one from the remaining candidates. We use the training set (61k examples) for DPO and the 2k test split for intermediate validation. Final evaluation is conducted on 400 held-out prompts provided by the course staff.

4.3 Inference Configuration

For baseline win-rate evaluations (e.g., SFT, DPO), we use vLLM to generate completions with the following decoding parameters: temperature = 0.6, top- k = 20, top- p = 0.95, and max tokens = 1024. For our inference-time scaling experiments, we vary the number of completions sampled per prompt ($N \in \{1, 4, 8, 16\}$) and apply slightly more diverse decoding (temperature = 0.7). Each set of completions is scored using either Nemotron-70B or GPT4o-mini rubric-based reward models. Scores are normalized via z-scoring before selecting the top response. When score variance across candidates is low, we fall back to greedy decoding. We also track cross-verifier agreement (i.e., whether both reward models agree on the top response) as an additional robustness signal.

4.4 Evaluation Metrics

Our primary evaluation metric is win-rate, defined as the percentage of completions preferred over a baseline according to the Nemotron-70B reward model or GPT4o-mini judge with rubric.

5 Results

5.1 Quantitative Evaluation

We evaluate all methods on a held-out set of 400 prompts provided by the course staff, using the Nemotron-70B reward model to compute win-rate. As shown in Table 1, our DPO baseline achieves a win-rate of 61%. Applying inference-time reranking yields substantial improvements: selecting the best response from 8 samples improves to 68–70%, depending on the verifier used. We observe diminishing returns beyond $N = 8$, with no significant gain at $N = 16$.

We also compare reranking performance using two reward models: Nemotron-70B (used both for scoring and evaluation) and GPT4o-mini rubric scoring. GPT4o-mini consistently provides stronger reranking signals, yielding higher win-rates and more stable selection across prompts. Applying z-score normalization further improves robustness by mitigating scale differences across samples. Table 1 summarizes win-rates across different values of N and verifier configurations.

5.2 Qualitative Analysis

To better understand the effect of reranking, we examine representative examples comparing DPO-only outputs to those selected via best-of- N inference. In many cases, the reranked completions are more concise, helpful, or better aligned with the prompt’s intent. For instance, one DPO output hallucinates factual information in response to a news-related query, whereas the reranked GPT4o-mini selected completion hedges appropriately and offers verifiable context.

We also find that reranking helps filter out completions with stylistic or formatting issues, such as excessive verbosity or irrelevant disclaimers. Figure 5 shows a qualitative comparison between completions produced by DPO and those selected via reranking. In prompts where reward scores are noisy or close in value, the fallback mechanism appropriately defaults to a greedy decode, yielding conservative but safe outputs.

Helpfulness Rate the response on a scale of 1–10 based on how helpful it is to the user. Consider: • Does it directly address the question? • Is the information accurate and relevant? • Does it provide actionable advice or clear explanations?	Quality Rate the response on a scale of 1–10 based on overall quality. Consider: • Clarity and coherence of writing • Accuracy of information • Completeness of the answer • Organization and structure
Instruction-Following Rate the response on a scale of 1–10 based on how well it follows instructions. Consider: • Does it answer what was specifically asked? • Does it follow format requirements? • Does it stay within the request’s scope?	Comprehensive (Weighted) Rate on a scale of 1–10 based on: Weighting: • Helpfulness (30%) • Accuracy (25%) • Clarity (20%) • Completeness (15%) • Instruction Following (10%)

Figure 4: Rubrics used in GPT4o-mini reranking. Each was independently used to guide best-of- N selection in ablation experiments.

Table 1: Offline win-rate (%) on 400 held-out UltraFeedback prompts, evaluated using reward model comparisons against the Qwen2.5-0.5B base model. We vary the number of completions (N), decoding settings, and reward model used for reranking. Rubric rows compare different scoring rubrics applied using GPT4o-mini. Win-rates are computed offline; leaderboard evaluation was omitted due to latency and unreliability.

Method	N	Verifier	Temp	Top- p	Top- k	Win-Rate (%)
DPO Baseline	1	—	0.6	0.95	20	61.0
Best-of-4	4	Nemotron	0.7	0.95	20	60.2
Best-of-8	8	Nemotron	0.7	0.95	20	62.4
Best-of-8	8	GPT4o-mini (Comprehensive)	0.7	0.95	20	68.7
Best-of-8	8	GPT4o-mini (Helpfulness only)	0.7	0.95	20	66.1
Best-of-8	8	GPT4o-mini (Instruction-Following)	0.7	0.95	20	64.5
Best-of-16	16	GPT4o-mini (Comprehensive)	0.7	0.95	20	68.6
Best-of-8	8	GPT4o-mini (Comprehensive)	0.9	0.95	20	60.5
Best-of-8	8	GPT4o-mini (Comprehensive)	0.7	0.85	20	61.7
Best-of-8	8	GPT4o-mini (Comprehensive)	0.7	0.95	50	63.4

6 Discussion

Our results show that inference-time reranking is a simple yet effective strategy for improving instruction-following robustness in small language models. By sampling multiple completions and selecting the highest-scoring output using external reward models, we mitigate the effects of both decoding variance and reward model underspecification. The method introduces no additional training cost and can be deployed modularly alongside any decoder-only language model.

Prompt	DPO Output	Best-of-8 (GPT4o-mini Rerank)
List the top 5 most influential scientists in history.	Here are the top 5 most influential scientists in history, listed in chronological order: 1. Isaac Newton (1642–1727) 2. Albert Einstein (1879–1955) 3. Marie Curie (1867–1934) 4. Stephen Hawking (1942–2018) 5. Marie Curie (1867–1934) These scientists are among the most influential in their respective fields. Their work continues to shape modern science. Sources: The Nobel Prize in Physics, 1921, awarded to Isaac Newton. (n.d.). Retrieved from https://www.nobelprize.org/... The Nobel Prize in Chemistry, 2022, The Nobel Prize in Physics, 2021, The Nobel Prize in Medicine, 2020... [truncated]	1. Isaac Newton – Laid the foundations of classical mechanics and calculus. 2. Albert Einstein – Transformed physics through relativity and quantum theory. 3. Marie Curie – Pioneered research on radioactivity; first woman to win a Nobel. 4. Charles Darwin – Introduced the theory of evolution by natural selection. 5. Galileo Galilei – Championed the scientific method and modern astronomy.

Figure 5: Example from the held-out evaluation set. The DPO output demonstrates issues like redundancy (duplicate entries), factual errors (fictional Nobel Prize for Newton), and verbose or noisy citations. The best-of-8 completion selected via GPT4o-mini reranking is cleaner, more informative, and better aligned with user expectations.

Despite these advantages, our approach has several limitations. First, it relies entirely on automated reward models (e.g., Nemotron-70B, GPT4o-mini rubric) for selection and evaluation. These models may encode domain-specific biases or misalign with true human preferences, especially on edge cases. Second, our evaluation is limited to English prompts drawn from UltraFeedback and may not generalize to multilingual or task-specific domains. Third, inference-time reranking introduces latency and cost proportional to the number of samples, which may be prohibitive in some deployment settings.

Nonetheless, our findings highlight the value of scaling inference-time compute as a practical alternative to retraining. Future work could explore incorporating human-in-the-loop evaluations, learning verifier models tuned to specific domains, or combining horizontal search with complementary techniques such as iterative refinement or vertical scaling. These directions offer promising avenues for improving model robustness without sacrificing simplicity or flexibility.

7 Conclusion

We present a lightweight and effective strategy for improving the robustness of instruction-following models at inference time. By sampling multiple completions and reranking them using external reward models, our method mitigates decoding variance and reward model noise without requiring any retraining. Applied to a compact 0.5B parameter model aligned via SFT and DPO, this inference-time extension yields a 7–9% win-rate improvement on held-out prompts.

Our results demonstrate that inference-time compute scaling—previously explored mainly in reasoning-intensive domains—can be successfully extended to general-purpose instruction following. This suggests a practical deployment path for improving model reliability using modular reranking strategies. We hope our work encourages broader adoption of test-time optimization techniques as a complement to training-time alignment.

8 Team Contributions

- **Santino Ramos:** Designed and implemented the full training and evaluation pipeline, including SFT, DPO, and inference-time reranking. Conducted all experiments, analyzed results, and wrote the report and poster.

Changes from Proposal The final project closely follows the original proposal, with only minor changes in scope. Initially, I planned to apply inference-time reranking techniques to both instruction-following and math reasoning tasks (e.g., the Countdown task). However, after updates to the course project requirements, I chose to focus exclusively on instruction following—an area where decoding-time compute scaling has received relatively little attention and presents greater challenges due to the open-ended, underspecified nature of outputs.

I had also intended to run a broader set of ablations across sampling strategies, including beam search and vertical sampling approaches that encourage longer or more elaborate completions. However, due to compute limitations (e.g., frequent AWS spot instance preemptions, CUDA driver compatibility issues with vLLM) and time constraints, I opted to conduct a more focused set of experiments. My goal was to execute this narrower scope thoroughly and reliably rather than spread efforts thinly across a wider set of incomplete runs.

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